

Most of the world' s farmed aquatic production is concentrated in East Asia & Pacific: evidence, definitions, and what can (and cannot yet) be verified

Executive summary (what the evidence supports directly)

1. **A rigorous, region-defined statement is strongly supported:** Using the FAO→World Bank→Our World in Data (OWID) "Aquaculture production" series (metric tons), **the World produced 126.94M metric tons in 2022**, and **East Asia & Pacific (World Bank region) produced 102.12M**—about **80%** of the world total ([1]).
 2. **"East Asia" (narrow) is not identical to "East Asia & Pacific (EAP)."** The sources used here define regions differently (World Bank regions in OWID/WDI; FAO regional groupings in SOFIA; "APAC" in Statista). Without an explicit crosswalk in the provided materials, the report treats **EAP** dominance as verifiable, while **"East Asia alone"** remains **not fully auditable from this evidence set** ([2]).
 3. **China is the core country-level driver by volume.** In OWID (2022), **China is listed at 75.44M metric tons**, far above Indonesia (14.78M) and India (10.24M) ([1]). In World Bank WDI (ER.FSH.AQUA.MT), China is listed at **78,276,941 metric tons (2023)** ([3]).
 4. **Global totals differ across sources because "aquaculture" can include algae and other non-animal products.** FAO SOFIA reports a **composite** total (animals + algae + small residual categories), while many "farmed fish" discussions implicitly mean animals only. SOFIA 2024 reports **130.9M tonnes in 2022** total aquaculture, split into **94.4M aquatic animals (live weight equivalent)** and **36.5M algae (wet weight)**, plus **2,700 tonnes shells & pearls** ([4]).
 5. **Forward-looking context points to continued aquaculture-led supply growth and Asia-led consumption growth**, but the provided OECD-FAO chapter page does **not** provide enough regional production tables to compute "future East Asia share" transparently ([5]).
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1) Scope and operational definitions (to prevent category-mismatch)

1.1 What "aquaculture production" means in the core quantitative series used here

This report uses **aquaculture production** as defined in OWID' s FAO-via-World Bank indicator documentation: **farming of aquatic organisms (fish, molluscs, crustaceans, aquatic plants)** with

production counting **outputs designated for final harvest for consumption; ornamental harvest is excluded** ([6]). Statista's framing is explicitly aligned on definition (aquaculture organisms + final harvest for consumption) in the material available from its page, although its country table values are paywalled ([7]).

Two interpretation rules follow from this definition:

- **"Aquaculture" here is broader than finfish.** It includes molluscs, crustaceans, and aquatic plants; claims about "farmed fish" based on this series are best read as **farmed aquatic production** unless an animals-only source is used ([6]).
- **"Production" is harvest output**, not installed capacity or biomass stocks—appropriate for "share of world production" comparisons ([6]).

1.2 Why FAO SOFIA totals don't always match OWID/WDI totals

FAO SOFIA explicitly reports a **composite** "world aquaculture production" concept combining:

- **Aquatic animals** (reported in **live weight equivalent**),
- **Algae** (reported in **wet weight**),
- **Shells and pearls** (small residual) ([4]).

This report therefore avoids mixing:

- **SOFIA composite totals** (animals + algae) with
- **animals-only notions of "farmed fish"** unless the mismatch is stated explicitly.

2) Regional concentration: what can be stated precisely and audited

2.1 The strongest auditable claim: EAP produces ~80% of world aquaculture output (2022, OWID/World Bank regions)

Using OWID's FAO→World Bank regional breakdown, in **2022**:

- **World: 126.94M metric tons**
- **East Asia & Pacific (EAP): 102.12M (~ 80%)** ([1]).

Other regions in the same OWID breakdown drop sharply in scale in 2022:

- **South Asia: 13.11M**
- **Latin America & Caribbean: 4.26M**

- **Europe & Central Asia: 4.01M**
- **Middle East, North Africa, Afghanistan & Pakistan: 1.95M**
- **Sub-Saharan Africa: 843.08k**
- **North America: 645.29k**

(all from the same OWID chart: [1]).

Boundary condition: This is a statement about **East Asia & Pacific as defined by the World Bank's region classification**, not "East Asia" narrowly.

 Image_36

2.2 Why "East Asia" ≠ "East Asia & Pacific (EAP)" (and why that matters for verification)

OWID's regional rollups follow **World Bank region groupings**. The World Bank publishes an explicit **East Asia & Pacific = 38 economies/territories** membership list (including China, Japan, Korea Rep., Indonesia, Viet Nam, Philippines, Australia, New Zealand, plus territories such as Hong Kong SAR, China; Macao SAR, China; Taiwan, China; Guam; American Samoa; Northern Mariana Islands) ([2]).

Implication for this topic: The headline "East Asia produces most of the world's farmed fish" can be verified **as written** only if "East Asia" is operationalized with a documented country list. In the provided materials, the defensible, auditable form is:

East Asia & Pacific (World Bank region) produces about 80% of global aquaculture production (2022, OWID/FAO via World Bank).

([1]; [2])

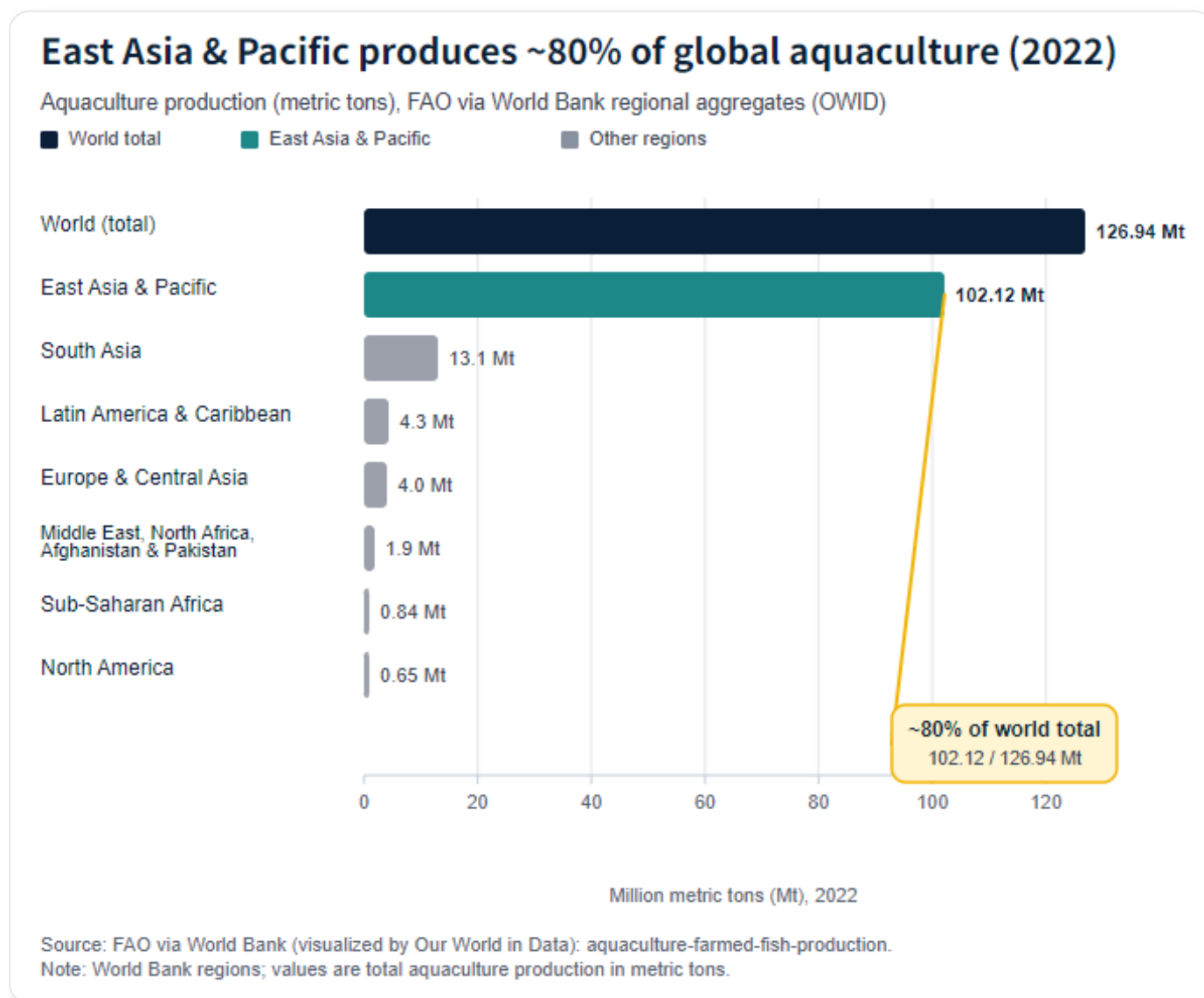
3) Country-level concentration: China and the next tier of producers

3.1 China's magnitude in open datasets (OWID 2022; WDI 2023; Statista APAC 2023 snippet)

- **OWID (2022):** China **75.44M metric tons**; Indonesia **14.78M**; India **10.24M**; Viet Nam **5.11M**; Bangladesh **2.73M**; Philippines **2.35M**; South Korea **2.31M** ([1]).
- **World Bank WDI** `ER.FSH.AQUA.MT` (2023 most recent shown for China): China **78,276,941 metric tons (2023)** ([3]).
- **Statista (APAC, 2023):** China **"over 78.3 million metric tons,"** with table values otherwise inaccessible in the provided extract due to paywall ([7]).

Interpretation note: The WDI page is explicit about units ("metric tons") and source attribution to FAO, but the excerpted metadata does not provide deeper methodological notes on aggregation or

category composition beyond provenance ([3]).



3.2 Benchmarks beyond China: large Asian producers plus non-Asian comparators (WDI examples)

The WDI country table provides scale comparisons across regions for the “most recent year” reported per economy (often 2023, but not uniformly). Examples visible in the provided material include:

- **Bangladesh: 2,852,572 (2023)**
- **Egypt, Arab Rep.: 1,552,430 (2023)**
- **Chile: 1,502,919 (2023)**
- **Ecuador: 1,234,088 (2023)**
- **Brazil: 793,096 (2023)**

([3]).

Year-alignment caveat (important for verification): WDI “Most Recent Year” varies by economy in the visible table (e.g., Channel Islands shown with **1,190 (2022)**), so cross-economy comparisons

should confirm the year, not assume all values are 2023 ([3]).

3.3 A concentrated “top producers” structure (SOFIA 2024) — reported, not re-computed here

FAO SOFIA 2024 reports that **10 countries account for “over 89.8%”** of total aquaculture output and lists: **China, Indonesia, India, Viet Nam, Bangladesh, the Philippines, Republic of Korea, Norway, Egypt, and Chile** ([4]).

Replication limit: The provided excerpt does not include the underlying country tonnages/shares used to compute “over 89.8%,” so this report treats it as a **FAO-reported concentration statistic** rather than a reproducible calculation from included tables.

3.4 Animals vs algae: China-led algae growth can change “aquaculture total” comparisons

SOFIA 2024 notes farmed algae increased from **35.1M tonnes (2020)** to **36.5M tonnes (2022)** (**+1.4M; +4.1%**) and that this growth was “**led by China,**” with other named contributors and offsets ([4]). This is one reason “aquaculture total” (including algae) can yield different country/region emphases than animals-only framing.

4) Subregional comparator: Southeast Asia’ s aquaculture-heavy production mix (SEAFDEC 2023)

To contextualize “Asia dominance” beyond China/East Asia, SEAFDEC’ s **Fisheries Statistics Summary 2023** reports Southeast Asia total production split into:

- **Aquaculture: 26,455,153 MT**
- **Marine capture: 18,885,858 MT**
- **Inland capture: 3,162,510 MT**

([8]).

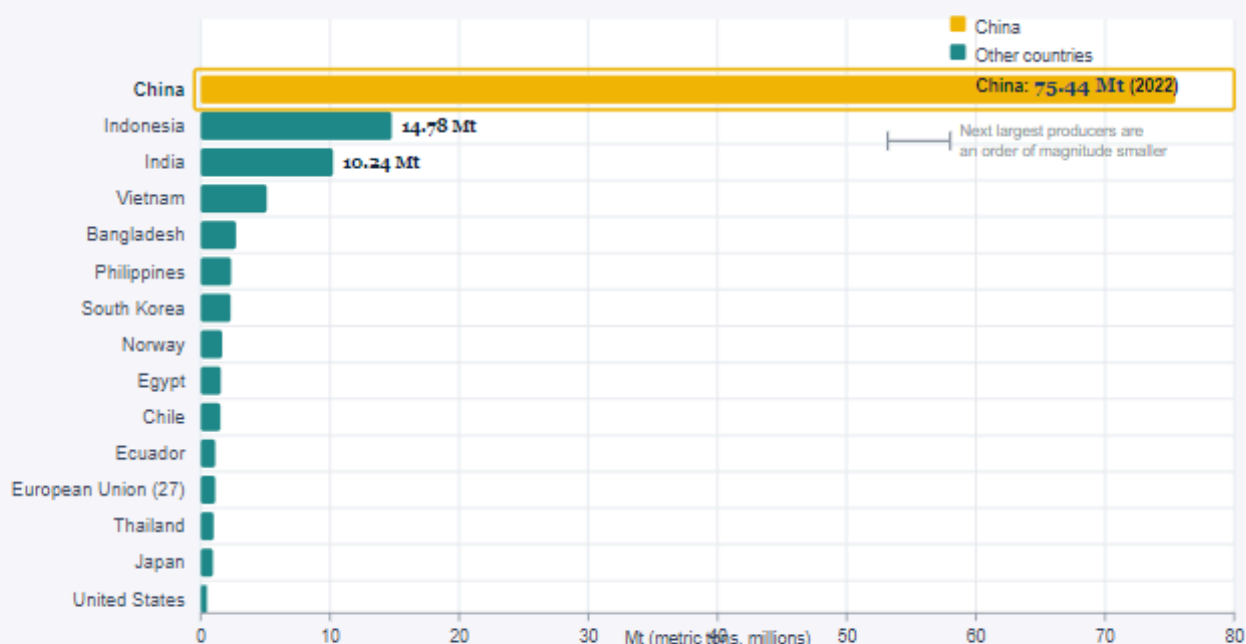
SEAFDEC also reports that (for 2023) aquaculture accounts for **~54.5% of quantity** and **~47.7% of value** in the region’ s total production mix ([9]).



SEAFDEC’ s country share breakdown further shows multi-polarity within Southeast Asia: by **2023 quantity, Indonesia** leads with **47.8%**, followed by **Viet Nam (19.3%)**, then **Myanmar (13.0%)** and **the Philippines (8.8%)** ([9]).

Aquaculture production by country (2022): China vs next-largest producers

OWID/FAO via World Bank country listing; values in metric tons (Mt). China's bar is highlighted for emphasis.



Source: Our World in Data (FAO via World Bank), Aquaculture production (metric tons), 2022 country values.

Note: Metric tons as reported in OWID aquaculture production dataset; cross-year comparisons should use consistent definitions.

How this supports the topic: Even without asserting a narrow “East Asia” percentage, the combination of (a) OWID’s EAP≈80% share and (b) SEAFDEC’s aquaculture-dominant Southeast Asia mix helps explain why “East Asia & Pacific” aggregates are so large: they span multiple high-output aquaculture systems, not only China.

5) Cross-source triangulation: what aligns cleanly, what does not, and what is missing

5.1 Alignment that *is* auditable from provided material

- **Concept definition alignment:** OWID’s definition and Statista’s stated definition both describe aquaculture as farmed aquatic organisms for final harvest for consumption; OWID explicitly excludes ornamental harvest ([6]; [7]).
- **Indicator identity:** World Bank labels the series as “**Aquaculture production (metric tons)**” with code `ER.FSH.AQUA.MT` and FAO attribution ([3]).
- **Region membership definition:** World Bank publishes the EAP membership list used to interpret OWID/WDI regional groupings ([2]).

5.2 Mismatches that require explicit handling (and cannot be “hand-waved”)

- **Region schemas differ:**
 - OWID regional totals rely on **World Bank regions** (EAP, South Asia, etc.) ([1]).
 - FAO SOFIA uses **FAO regional groupings** (e.g., “Asia” as a region) when attributing growth shares ([4]).
 - Statista’s **APAC** grouping is separate again, and its table is not accessible in the provided extract ([10]).

Consequence: EAP shares cannot be algebraically combined with FAO “Asia” growth attributions without a crosswalk.
- **Category composition differs:** SOFIA’s composite totals (animals + algae) are not necessarily identical to other “aquaculture production” totals unless category scope is matched ([4]).

5.3 High-value evidence that is referenced but not present (therefore not used for numeric claims here)

SOFIA 2024 clearly points to the existence of key audit tables:

- **Table 2:** “World and regional aquaculture production and growth” (listed around p. 15)
 - **Table 3:** “World aquaculture production of aquatic animals by region and selected major producers” (listed around p. 17)
- ([11]).

However, the provided excerpts do **not** include the numeric cell values from these tables. This report therefore uses SOFIA for the **global totals and category splits explicitly quoted in the excerpt**, and treats Table 2/3 as **locators for future extraction**, not as numeric evidence in this write-up.

Licensing note relevant to reproducibility: SOFIA 2024 is **CC BY 4.0**, but the DOI page emphasizes that some third-party materials (tables/figures/images) and photographs may require additional permissions ([11]; [12]).

6) Supporting context: production systems, spatial measurement, and planning lenses (kept distinct from the core share claim)

6.1 System composition matters for interpreting “dominance” (SOFIA 2022 animals breakdown)

SOFIA 2022 provides a detailed animals-only breakdown (2020) showing why aquaculture dominance is not about a niche category:

- **Total aquaculture (2020): 122.6M tonnes**, comprising **87.5M aquatic animals, 35.1M algae, 700 tonnes shells & pearls**; farm-gate value **USD 281.5B** ([13]).

- **Aquatic animals composition (2020):** finfish **57.5M tonnes** (~66%), molluscs **17.7M**, crustaceans **11.2M**, aquatic invertebrates **525,000**, semi-aquatic species **537,000** ([13]).
- **Finfish environment split (2020):** inland **49.1M tonnes** vs mariculture/sea & coastal on-shore **8.3M** ([13]).

Why this is relevant here: When readers interpret “East Asia & Pacific dominates aquaculture,” SOFIA 2022 clarifies that the dominated object is heavily finfish and heavily inland—system types that have distinct land/water constraints and policy levers.

6.2 Observed spatial footprint (remote sensing) complements production statistics—but answers a different question

A Southeast Asia remote-sensing study (1990–2020) uses Landsat and Sentinel-2 in Google Earth Engine and reports mapping accuracy **>89.53% up to 95.49%**, alongside a redistribution of aquaculture area shares toward Vietnam over time ([14]). This does **not** directly measure production tonnage, but it strengthens interpretation of how aquaculture footprints can shift within Asia even when Asia remains dominant globally.

6.3 Alternative spatial taxonomies exist (FAO Major Fishing Areas), but linkage to aquaculture totals is not established here

FAO publishes Major Fishing Areas reference materials (PDFs and a shapefile endpoint), which could support GIS-based aggregation frameworks, but the provided materials do not demonstrate an operational join between those areas and aquaculture production statistics (e.g., via FishStatJ/FAOSTAT tables) ([15]; example area page [16]; shapefile endpoint [17]).

6.4 Planning frameworks illustrate “where expansion could occur” (Nigeria example), but do not change the current concentration finding

A University of Stirling PhD thesis (2022) develops scenario-driven spatial decision-support for aquaculture expansion planning in Nigeria to 2035 using **Delphi → Scenario Analysis + SMCE + GIS**, and reports a related peer-reviewed paper: Yakubu et al. (2022) in *Aquaculture Reports* ([18]). This is methodologically relevant to “how to plan/locate aquaculture,” but it is not evidence about East Asia/EAP shares.

Aquaculture concentration and FAO reference context

Schematic summary of share estimates, FAO Major Fishing Areas assets, SE Asia mapping results, and SOFIA 2022 definitions.

Asia aquaculture share

~90%

Asia hosts ~90% of the world's aquaculture activities.
Attribution: SE Asia remote-sensing mapping paper citing Handside et al., 2006.

FAO Major Fishing Areas assets

27 fact sheets

- Maps/Poster PDF
- Detailed areas list PDF
- Shapefile (GeoNetwork UUID: ac02a460_da52_11dc_9d70_00177293bd28)

Taxonomy note: 27 fact sheets; inland 01-08, marine 18-88.

SE Asia remote-sensing mapping

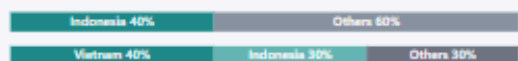
Accuracy range reported in the study.

>89.53% 95.49%

Distribution shift:

Pre-period: Indonesia / Other = 40% / 60%

Post-period: Vietnam / Indonesia / Other = 40% / 30% / 30%



NW shift

FAO SOFIA definitional split

130.9 Mt (2022)

94.4 Mt aquatic animals (live weight eq.) + 36.5 Mt algae (wet weight)
+ 2,700 t shells/pearls

Finfish 57.5 Mt (2020)

Inland 49.1 + Marine/coastal 8.3

Units differ (live weight vs wet weight).

⚠ Evidence gap: not confirmed here whether FAO/FAO/FAO provide aquaculture by FAO Major Fishing Area, or whether the MFA shapefile attributes enable automated collapse to countries/regions.

7) Forward-looking implications (bounded to what is supported in the provided materials)

7.1 Supply: aquaculture is projected to provide most incremental growth (OECD-FAO)

OECD-FAO projects total fisheries & aquaculture production reaching **212 Mt by 2034**, with aquaculture providing **“over 85%”** of additional output and reaching **56%** of total production by 2034 ([5]). Separately, the same chapter page reports a 2024 snapshot of global production at about **193 Mt** ([5]).

What this does (and does not) imply for “East Asia share” : It supports a general mechanism—future supply growth is aquaculture-led—but does not, on its own, quantify where that aquaculture growth occurs by region in a way that can be mapped to World Bank EAP or “East Asia.”

7.2 Demand: consumption growth is mostly attributed to Asia (OECD-FAO)

OECD-FAO projects global per-capita consumption rising to **21.8 kg by 2034** (from **21.1 kg in 2022–2024**) and attributes **75% of consumption growth** to **Asia** (Africa 15%, Americas 11%, Oceania 1%, Europe slightly declining) ([5]). It also states that by 2034 **Asia accounts for 73% of aquatic foods available for consumption**, with **China alone at 37%** ([5]).

7.3 Why the report does not compute a “2034 East Asia production share”

The OECD-FAO chapter page notes that detailed projections are in an online Statistical Annex, but those annex tables are not included in the provided material; therefore, any computed “East Asia (or EAP) share in 2034” would not be verifiable from this evidence set ([5]).

Conclusion (tight, source-bounded)

- The evidence provided supports a clear, verifiable core finding: **East Asia & Pacific (World Bank region) accounts for ~80% of global aquaculture production (metric tons) in 2022** in the OWID/FAO-via-World Bank series ([1]).
 - At the country level, **China’s production scale** (OWID 2022: **75.44M**; WDI 2023: **78,276,941**) makes it the dominant single contributor to both Asia/EAP totals and global concentration patterns ([1]; [3]).
 - Stronger statements about “**East Asia (narrow) share**” require a documented country list and a crosswalk among **World Bank EAP**, **FAO regions**, and **Statista APAC**—which are not provided here; accordingly, this report treats EAP dominance as the auditable geographic claim and keeps “East Asia” language explicitly qualified ([2]).
 - Differences between **SOFIA composite totals (animals + algae)** and other “aquaculture production” totals are not errors; they reflect category scope and measurement conventions that must be matched before numerical reconciliation ([4]).
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References (curated to sources actually used above)

- Our World in Data (FAO via World Bank): Aquaculture production chart + sources/processing: [1] and [6]
- World Bank WDI indicator `ER.FSH.AQUA.MT` : [3]
- World Bank “Country and Lending Groups” (region membership lists): [2]
- FAO SOFIA 2024 (DOI landing page + aquaculture production chapter): [11] and [4]
- FAO SOFIA 2022 aquaculture production chapter: [13]
- SEAFDEC Fisheries Statistics Summary 2023: [8] and category landing: [9]
- OECD-FAO Agricultural Outlook 2025–2034 (Fish and other aquatic products chapter): [5]
- Statista APAC aquaculture production page (definition + paywalled table context): [7]
- Remote sensing study (Southeast Asia aquaculture footprint): [14]
- Yakubu et al. (2022) Aquaculture Reports paper: [18]

References

1. [OWID chart](#)
2. [World Bank region list](#)
3. [WDI indicator page](#)

4. [FAO SOFIA 2024 aquaculture production](#)
5. [OECD-FAO Outlook 2025–2034 chapter](#)
6. [OWID sources & processing](#)
7. [Statista APAC definition page](#)
8. [SEAFDEC Stat 2023](#)
9. [SEAFDEC Stat 2023](#)
10. [Statista APAC page](#)
11. [SOFIA 2024 DOI landing page](#)
12. [CC BY 4.0 legal code](#)
13. [SOFIA 2022 aquaculture production](#)
14. [ScienceDirect article](#)
15. [FAO Major Fishing Areas concept page](#)
16. [FAO Area 27](#)
17. [FAO GeoNetwork UUID](#)
18. [Aquaculture Reports paper](#)